Lab No: 36 Date: 2026/03/26

Title: Configuring SSL-Enabled Apache (HTTPS) Server (self-signed)

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a) Install required package for HTTPs server

```
aregmi@vbox:~
aregmi@vbox:~$ sudo yum install httpd mod_ssl openssl -y
[sudo] password for aregmi:
Last metadata expiration check: 1:01:13 ago on Sun 22 Mar 2026 09:28:16 AM +0545
.
Package openssl-1:3.5.5-1.el10.x86_64 is already installed.
Dependencies resolved.
================================================================================
 Package                 Arch        Version                Repository      Size
================================================================================
Installing:
 httpd                   x86_64      2.4.63-13.el10         appstream       49 k
 mod_ssl                 x86_64      1:2.4.63-13.el10       appstream      110 k
Installing dependencies:
 apr                     x86_64      1.7.5-3.el10           appstream      127 k
 apr-util                x86_64      1.6.3-23.el10          appstream       97 k
 apr-util-lmdb           x86_64      1.6.3-23.el10          appstream       13 k
 centos-logos-httpd      noarch      100.4-1.el10           appstream      854 k
 httpd-core              x86_64      2.4.63-13.el10         appstream      1.5 M
 httpd-filesystem        noarch      2.4.63-13.el10         appstream       14 k
 httpd-tools             x86_64      2.4.63-13.el10         appstream       83 k
Installing weak dependencies:
 apr-util-openssl        x86_64      1.6.3-23.el10          appstream       15 k
 mod_http2               x86_64      2.0.29-3.el10          appstream      164 k
 mod_lua                 x86_64      2.4.63-13.el10         appstream       60 k
```

b) Allow https (443) packets to enter through firewall

```
aregmi@vbox:~
aregmi@vbox:~$ sudo firewall-cmd --permanent --add-service=https
success
aregmi@vbox:~$ sudo firewall-cmd --reload
success
```

c) Start R enable service

```
aregmi@vbox:~$ sudo systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/
usr/lib/systemd/system/httpd.service'.
aregmi@vbox:~$ sudo systemctl start httpd
aregmi@vbox:~$ sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
> -keyout /etc/pki/tls/private/selfsigned.key \
> -out /etc/pki/tls/certs/selfsigned.crt
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```
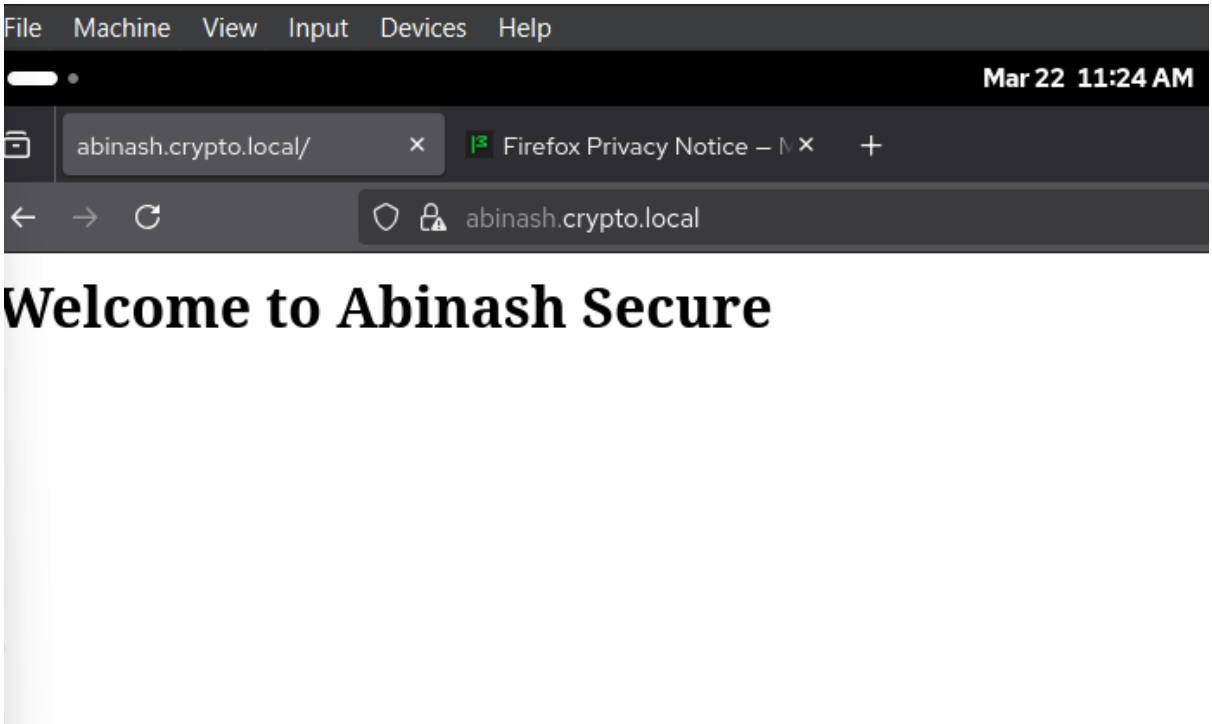

d) Generate self-signed key and cert files using OpenSSL

```
aregmi@vbox:~$ sudo vi /etc/httpd/conf.d/ssl.conf

No manual entry for SSLCertificateFile

shell returned 16

Press ENTER or type command to continue
No manual entry for SSLCertificateFile

shell returned 16

Press ENTER or type command to continue
No manual entry for SSLCertificateFile

shell returned 16

Press ENTER or type command to continue
aregmi@vbox:~$ sudo vi /var/www/html/index.html
[sudo] password for aregmi:
aregmi@vbox:~$ sudo vi /etc/hosts
aregmi@vbox:~$ sudo systemctl restart httpd
```

e) Configure web server to listen from port 443 and set Document and open webpage named abinash.crypto.local